

Campus Enterprise Network Design (Cisco Packet Tracer)

Overview

Designed a scalable multi-campus network for SZABIST Karachi connecting six campuses using a hierarchical architecture. The network ensures efficient communication, segmentation, and security.

Architecture

- Core–Distribution–Access layered design
 - Edge router in 99 Campus connecting all buildings
 - VLAN-based segmentation
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Key Technologies

- VLANs for segmentation
 - VLSM for IP optimization
 - OSPF dynamic routing
 - DHCP server (154 Campus)
 - Static IPs for servers/printers
 - ACLs for traffic control
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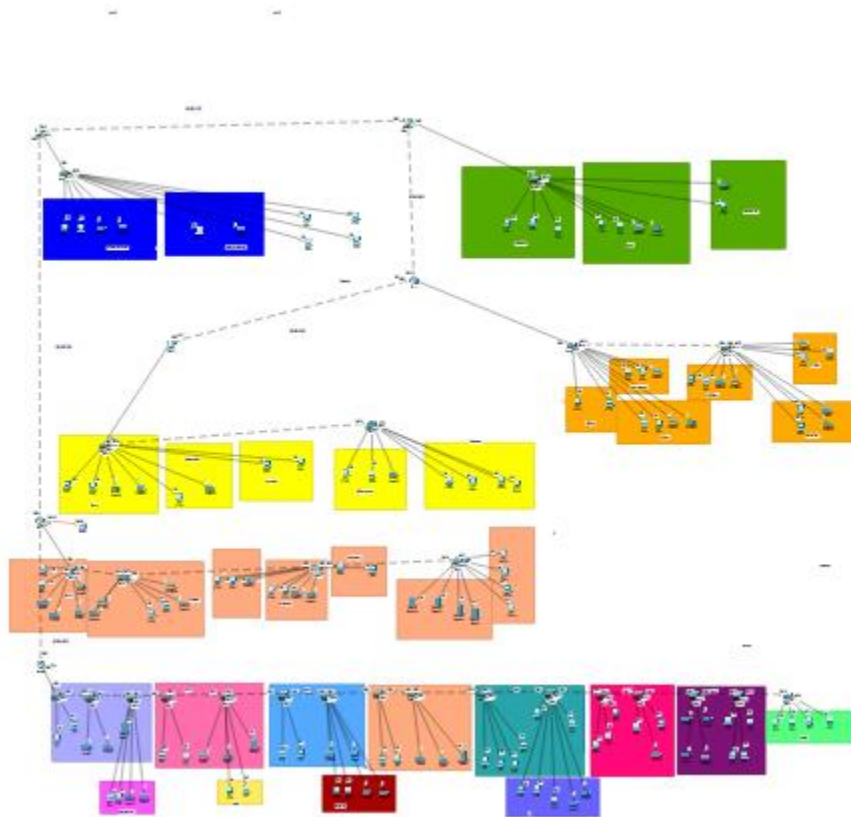
Network Scale

- 6 campuses
 - Multiple departments
 - Hundreds of connected devices
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Key Achievements

- Efficient IP allocation using VLSM
 - Reduced broadcast traffic using VLANs
 - Built scalable enterprise-level network
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Topology



Tools Used

Cisco Packet Tracer

Future Improvements

- Redundancy (HSRP, STP)
- Network monitoring
- Advanced security